01

Revogrocers Sales Performance Analysis

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Business Context

Company Overview

- RevoGrocers is a fictional grocery retail business that operates across multiple locations, providing a wide range of grocery products to customers.
- The company aims to optimize sales strategies, improve customer experience, and maximize revenue using data-driven insights.

Objectives

- Identify top-performing product categories
- Analyze factors driving revenue (units sold vs. customer count)
- Evaluate pricing strategies and their impact on sales

Disclaimer

- This analysis is based on a public dataset from Kaggle: 🔗 Kaggle Grocery Sales Dataset
- RevoGrocers is a fictional company created for analytical purposes.
- Any insights or recommendations are based on this dataset and do not reflect real-world data.

Link querry :
https://console.cloud.google.com/bigquery?ws=!1m7!1m6!12m5!1m3!1srevou-sql-class-479302!2sus-central1!3s8528670a-bc07-471b-a376-2c8b73b81d94!2e1

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Dataset Overview

Key Tables	Relevant Fields
categories	1. categoryid 2. categoryname
sales	1. SalesID 2. CustomerID 3. ProductID 4. Quantity 5. Discount 6. TotalPrice 7. SalesDate 8. TransactionNumber
products	1. Productid 2. Productname 3. Price 4. Categoryid
customers	1. Customerid 2. firstname 3. middleinitial 4. lastname

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1. Identify the product category that generates the highest revenue after discount.

Row	ProductCategory	RevenueAfterDis...
1	Confections	556930717.3468...
2	Meat	492888844.6946...
3	Poultry	440025564.9656...
4	Cereals	427393431.9101...
5	Snails	372084885.2075...
6	Beverages	366515024.0050...
7	Produce	362861133.5171...
8	Dairy	354358156.7688...
9	Seafood	330527987.4676...
10	Grain	323879126.6500...
11	Shell fish	299598294.0299...

```sql
SELECT
  categoryname AS ProductCategory,
  SUM((Quantity * Price) - (Discount * Quantity * Price)) AS RevenueAfterDiscount
FROM `fsda-sql-01.grocery_dataset.sales` AS sales
INNER JOIN `fsda-sql-01.grocery_dataset.products` AS products
  ON sales.ProductID = products.productid
INNER JOIN `fsda-sql-01.grocery_dataset.categories` AS categories
  ON products.categoryid = categories.categoryid
GROUP BY ProductCategory
ORDER BY RevenueAfterDiscount DESC;
```

The highest revenue-generating category is Confections, contributing 12% to total revenue. This indicates strong customer demand for Confections, and potential strategies could involve increasing product variety or targeted promotions.

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2. Assess the relation between revenue after discount and total units sold for each product category.

Row	ProductCategory	RevenueAfterDis...	TotalUnitsSold
1	Confections	556930717.3468...	11078474
2	Meat	492888844.6946...	9719292
3	Poultry	440025564.9656...	9159847
4	Cereals	427393431.9101...	8735296
5	Snails	372084885.2075...	7199409
6	Beverages	366515024.0050...	7393693
7	Produce	362861133.5171...	8174673
8	Dairy	354358156.7688...	6815143
9	Seafood	330527987.4676...	6996152
10	Grain	323879126.6500...	5433152
11	Shell fish	299598294.0299...	6983457

```sql
SELECT
  categoryname AS ProductCategory,
  SUM((Quantity * Price) - (Discount * Quantity * Price)) AS RevenueAfterDiscount,
  SUM(Quantity) AS TotalUnitsSold
FROM `fsda-sql-01.grocery_dataset.sales` AS sales
INNER JOIN `fsda-sql-01.grocery_dataset.products` AS products
  ON sales.ProductID = products.productid
INNER JOIN `fsda-sql-01.grocery_dataset.categories` AS categories
  ON products.categoryid = categories.categoryid
GROUP BY ProductCategory
ORDER BY RevenueAfterDiscount DESC;
```

Overall, there is a positive correlation between revenue and total units sold: category with higher revenue generally also have higher total units sold.

Two anomalies (Produce & Shell fish) which show relatively high units sold compared to the revenue, indicating it is volume-driven with lower average price per unit or higher discounts.

3. Find the relation between revenue after discount and the number of unique customers for each product category."

Row	ProductCategory	RevenueAfterDis...	NumberofUnique...
1	Confections	556930717.3468...	98743
2	Meat	492888844.6946...	98701
3	Poultry	440025564.9656...	98679
4	Cereals	427393431.9101...	98651
5	Snails	372084885.2075...	98376
6	Beverages	366515024.0050...	98424
7	Produce	362861133.5171...	98601
8	Dairy	354358156.7688...	98308
9	Seafood	330527987.4676...	98334
10	Grain	323879126.6500...	97335
11	Shell fish	299598294.0299...	98338

```sql
SELECT
  categoryname AS ProductCategory,
  SUM((Quantity * Price) - (Discount * Quantity * Price)) AS RevenueAfterDiscount,
  COUNT(DISTINCT CustomerID) AS NumberofUniqueCustomers
FROM `fsda-sql-01.grocery_dataset.sales` AS sales
INNER JOIN `fsda-sql-01.grocery_dataset.products` AS products
  ON sales.ProductID = products.productid
INNER JOIN `fsda-sql-01.grocery_dataset.categories` AS categories
  ON products.categoryid = categories.categoryid
GROUP BY ProductCategory
ORDER BY RevenueAfterDiscount DESC;
```

No. relation between revenue and number of unique customer

Revenue differences are not driven by customer count, since the number of unique customers is around the same.

The differences in revenue across categories are more influenced by:

- Average spending per customer
- Discount intensity (categories with larger discounts will have lower revenue even if same customers count).

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4. Calculate the average price unit for each product category in catalogue.

Row	ProductCategory	AvgUnitPrice
1	Grain	61.40394642857...
2	Dairy	53.55683142857...
3	Snails	53.19933243243...
4	Meat	52.27465199999...
5	Confections	51.84993157894...
6	Beverages	51.04368421052...
7	Cereals	50.41638888888...
8	Poultry	49.46088297872...
9	Seafood	48.66665555555...
10	Produce	45.79683333333...
11	Shell fish	44.27278333333...

```sql
SELECT
  categoryname AS ProductCategory,
  AVG(p.Price) AS AvgUnitPrice
FROM `fsda-sql-01.grocery_dataset.products` AS p
INNER JOIN `fsda-sql-01.grocery_dataset.categories` AS c
  ON p.categoryid = c.categoryid
GROUP BY ProductCategory
ORDER BY AvgUnitPrice DESC;
```

- Grain has the highest unit price (61.40) but relatively low revenue & total unit sold → an expensive product but low sales volume.
- Shell fish has the lowest unit price (44.27) & low revenue but relatively high units sold compared to the revenue → meaning it is volume-driven, purchased frequently because cheap price.
- Confections and Meat are in the middle, suitable for a mass-market strategy with moderate margins.
- Produce also has a low price (45.77) & relatively high units sold compared to the revenue → meaning it is volume-driven

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5. Evaluate the relation between the average price per unit and the number of buyers (unique customers) per category.

Row	ProductCategory	AvgUnitPrice	NumberofUnique...
1	Grain	61.43254359308...	97335
2	Dairy	53.61147548203...	98308
3	Snails	53.28068015957...	98376
4	Meat	52.31211525017...	98701
5	Confections	51.81190322273...	98743
6	Beverages	51.12591015803...	98424
7	Cereals	50.43802735917...	98651
8	Poultry	49.50921884342...	98679
9	Seafood	48.67796017344...	98334
10	Produce	45.78994061087...	98601
11	Shell fish	44.23266616321...	98338

```sql
SELECT
  c.categoryname AS ProductCategory,
  AVG(p.price) AS AvgUnitPrice,
  COUNT(DISTINCT s.CustomerID) AS NumberofUniqueCustomers
FROM `fsda-sql-01.grocery_dataset.sales` AS s
INNER JOIN `fsda-sql-01.grocery_dataset.products` AS p
  ON s.ProductID = p.productid
INNER JOIN `fsda-sql-01.grocery_dataset.categories` AS c
  ON p.categoryid = c.categoryid
GROUP BY ProductCategory
ORDER BY AvgUnitPrice DESC;
```

- No relation between Average Price per Unit & Number of Unique Customers.
- Number of Unique customers are around the same value (97–98k) across categories, regardless of average unit price.
- Price level does not significantly affect customer reach.
- Revenue are therefore explained by spending intensity per customer (Total Unit Sold), not by customer count.

6. Which categories contribute the most to overall revenue after discount (percentage-wise)?

Row	ProductCategory	RevenueAfterDis...	RevenuePercenta...
1	Confections	556930717.3468...	12.87
2	Meat	492888844.6946...	11.39
3	Poultry	440025564.9656...	10.17
4	Cereals	427393431.9101...	9.88
5	Snails	372084885.2075...	8.6
6	Beverages	366515024.0050...	8.47
7	Produce	362861133.5171...	8.39
8	Dairy	354358156.7688...	8.19
9	Seafood	330527987.4676...	7.64
10	Grain	323879126.6500...	7.48
11	Shell fish	299598294.0299...	6.92

```sql
SELECT
  c.categoryname AS ProductCategory,
  SUM((s.Quantity * p.Price) - (s.Discount * s.Quantity * p.Price)) AS RevenueAfterDiscount,
  ROUND(
    SUM((s.Quantity * p.Price) - (s.Discount * s.Quantity * p.Price))
    / SUM(SUM((s.Quantity * p.Price) - (s.Discount * s.Quantity * p.Price))) OVER() * 100, 2
  ) AS RevenuePercentage
FROM `fsda-sql-01.grocery_dataset.sales` AS s
INNER JOIN `fsda-sql-01.grocery_dataset.products` AS p
  ON s.ProductID = p.productid
INNER JOIN `fsda-sql-01.grocery_dataset.categories` AS c
  ON p.categoryid = c.categoryid
GROUP BY ProductCategory
ORDER BY RevenuePercentage DESC;
```

The highest revenue-generating category is Confections, contributing 12% to total revenue. This indicates strong customer demand for Confections, and potential strategies could involve increasing product variety or targeted promotions.

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7. Which product categories have the highest repeat purchase rate?

Row	ProductCategory	RepeatPurchase...
1	Confections	99.85315414763...
2	Meat	99.61195935198...
3	Poultry	99.43554353003...
4	Cereals	99.20527921663...
5	Produce	98.93408788957...
6	Beverages	98.22705844103...
7	Snails	97.91412539643...
8	Seafood	97.76679480139...
9	Shell fish	97.67739836075...
10	Dairy	97.32371729665...
11	Grain	93.68058766117...

```sql
WITH customer_category_counts AS (
  SELECT
    c.categoryname AS ProductCategory,
    s.CustomerID,
    COUNT(DISTINCT s.SalesID) AS PurchaseCount
  FROM `fsda-sql-01.grocery_dataset.sales` AS s
  INNER JOIN `fsda-sql-01.grocery_dataset.products` AS p
    ON s.ProductID = p.productid
  INNER JOIN `fsda-sql-01.grocery_dataset.categories` AS c
    ON p.categoryid = c.categoryid
  GROUP BY ProductCategory, CustomerID
)
SELECT
  ProductCategory,
  COUNT(DISTINCT CASE WHEN PurchaseCount >= 2 THEN CustomerID END)
    / COUNT(DISTINCT CustomerID) * 100 AS RepeatPurchaseRate
FROM customer_category_counts
GROUP BY ProductCategory
ORDER BY RepeatPurchaseRate DESC;
```

Repeat purchase behavior is extremely strong across all categories, with all exceeding 90%. This suggests high customer loyalty or habitual buying patterns

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8. Find the cumulative amount of transaction of the top user (user with highest transaction value, window function)

TopUser	CumulativeTrans...	ContributionPerc...	RankUser
Wayne L Chan	130324.1589599...	0.003	1
Olivia K Dean	123004.8642499...	0.0028	2
Ronda U Wallace	121922.59325	0.0028	3
Ericka H O'Connor	121883.9525000...	0.0028	4
Paula H Lin	120849.7785000...	0.0028	5
Benny D Wilson	119853.83904	0.0028	6
Tamiko D Newman	119418.6457500...	0.0028	7
Kerri I Bautista	119180.3025000...	0.0028	8
Jami N York	118687.7397500...	0.0027	9
Roberto D Durham	118624.5844999...	0.0027	10

```sql
WITH user_total AS (
  SELECT
    s.CustomerID,
    SUM((s.Quantity * p.Price) - (s.Discount * s.Quantity * p.Price)) AS CumulativeTransaction
  FROM `fsda-sql-01.grocery_dataset.sales` AS s
  INNER JOIN `fsda-sql-01.grocery_dataset.products` AS p
    ON s.ProductID = p.productid
  GROUP BY s.CustomerID
)
SELECT
  CONCAT(c.firstname, ' ', COALESCE(c.middleinitial, ''), ' ', c.lastname) AS TopUser,
  u.CumulativeTransaction,
  ROUND(
    u.CumulativeTransaction / SUM(u.CumulativeTransaction) OVER() * 100, 4
  ) AS ContributionPercentage,
  ROW_NUMBER() OVER (ORDER BY u.CumulativeTransaction DESC) AS RankUser
FROM user_total u
JOIN `fsda-sql-01.grocery_dataset.customers` AS c
  ON u.CustomerID = c.customerid
ORDER BY RankUser
LIMIT 10;
```

- Even the highest spender contributes only 0.003% of total revenue
- Revenue is widely distributed among customers
- No dominant customers
- Revenue depends on many customers, not a few Top Users.
- Focus on retention & repeat purchase, example : Loyalty program

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9. After executing the SQL queries, summarize your overall findings regarding the sales performance across product categories.

1. The highest revenue-generating category is Confections, contributing 12% of total revenue.
2. There is a positive correlation between Revenue and Total Units Sold: category with higher revenue generally also have higher total units sold.
3. No relation between Revenue and Number of Unique Customers.
4. No relation between Average Price per Unit and Number of Unique Customers.
5. Repeat Purchase behavior is extremely strong (>90%) across all categories
6. Revenue is widely distributed among customers. No dominant Top Users

Premium Category	Grain	expensive price - low sales volume
Mass-market categories	Shell fish & Produce	cheap price – high sales volume
Middle-tier categories	Confections & Meat	middle price – high sales volume

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Summary of Key Findings

- Top revenue-generating category is Confections (12% of total revenue)
- Positive correlation between Revenue & Total Units Sold
- No relation between Revenue and Number of Unique Customers
- Pricing does not significantly impact customer behavior
- Shell fish & Produce has low price & relatively high units sold compared to the revenue

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Recommendation):

Insights / Observation	Business Impact	Isolation	Prioritization	Recommendation
Top revenue-generating category is Confections (12% of total revenue)	Prioritizing to invest more in high-performing categories can boost revenue	Confections category	High High impact – High effort – Many data point	1. Increasing product variety of this category 2. Push targeted promotion to boost sales volume & repeat purchase 3. Push marketing to gain new customer in this category
Positive correlation between Revenue & Total Units Sold	Focus on increasing Total Unit Sold can boost revenue	Order size (quantity per order)	High High impact – High effort – Many data point	1. Discount tier, cashback, free shipping, point system for bulk order only 2. Bundling
No relation between Revenue and Number of Unique Customers	Focus on existing customer rather than gaining new customers	1. Average spending customer 2. Retention	High High impact – High effort – Many data point	1. Loyalty program 2. Gamification apps 3. Twin date campaign & special events to entertain existing customer

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RECOMMENDATION):

Insights / Observation	Business Impact	Isolation	Prioritization	Recommendation
Repeat Purchase behavior is extremely strong (>90%) across all categories	Focus on increasing Repeat Purchase can boost revenue	Repeat Purchase Rate	Medium Medium impact – Medium effort – Many data point	1. Product recommendation on landing page based on customer segmentation & history 2. Loyalty program 3. Bundling high repeat products 4. Email / WA followup 5. Push customer review
Revenue is widely distributed among customers. No dominant Top Users	Focus on overall customer experience rather than finding VIP customers	Customer experience	Low Low impact – Medium effort – Many data point	1. Optimize checkout flow 2. Reduce app crash during peak shopping hours
Pricing does not significantly impact customer behavior	Pricing does not significantly impact revenue	Pricing strategy	Low Low impact – High effort – Many data point	1. No need to modify pricing strategy 2. No need extensive discount to retain existing customers or gain new customers

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Recommendation):

Insights / Observation	Business Impact	Isolation	Prioritization	Recommendation
Premium categories (expensive price - low sales volume) → Grain	Focus on margin, quality, & differentiation	Grain category	Medium Medium impact – Medium effort – Medium data point	1. Focus on quality & differentation 2. High margin strategy
Middle-tier categories (middle price – high sales volume) → Confections, Meat	Focus on volume & margin	Confections & Meat category	Medium Medium impact – Medium effort – Medium data point	1. Increase margin on high volume, high repeat category 2. Push promotion to attract new customer in this category
Mass-market categories (cheap price – high sales volume) → Produce, Shell fish	Focus on volume, distribution, frequency	Produce & Shell fish category	Medium Medium impact – Medium effort – Medium data point	1. Discount tier, cashback, free shipping, point system for bulk order only 2. Bundling

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> Revogrocers can improve profitability and customer engagement with invest more in high-performing categories, increasing quantity per order, average spending customer, repeat purchase, customer experience while maintaining retention rate.

— Final Business Takeaway

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THANK YOU!

Do you have any questions?

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